

SMART WASTE BIN

Introduction

Waste management is a growing challenge especially in modern public spaces, shopping malls, shared accommodations, commercial buildings, and indoor social events where littering often occurs due to overflowing bins, irregular waste collection schedules or poor hygiene practices. Traditional waste bins require frequent manual checks to determine if they are full, and open-lid bins can attract pests and spread germs.

To tackle this issue, I am developing a Smart Waste Bin, an IoT-enabled solution designed to improve waste disposal efficiency. This bin automatically monitors its fill level and features an intelligent lid mechanism that only opens when someone intends to dispose of waste.

Problem Statement

1. No fill indicator in traditional bins leads to overflow.
2. Bins with simple motion sensors waste energy, opening anyhow and spreading bad smells.
3. Waste collectors work blindly since there's no real-time data on bin levels.

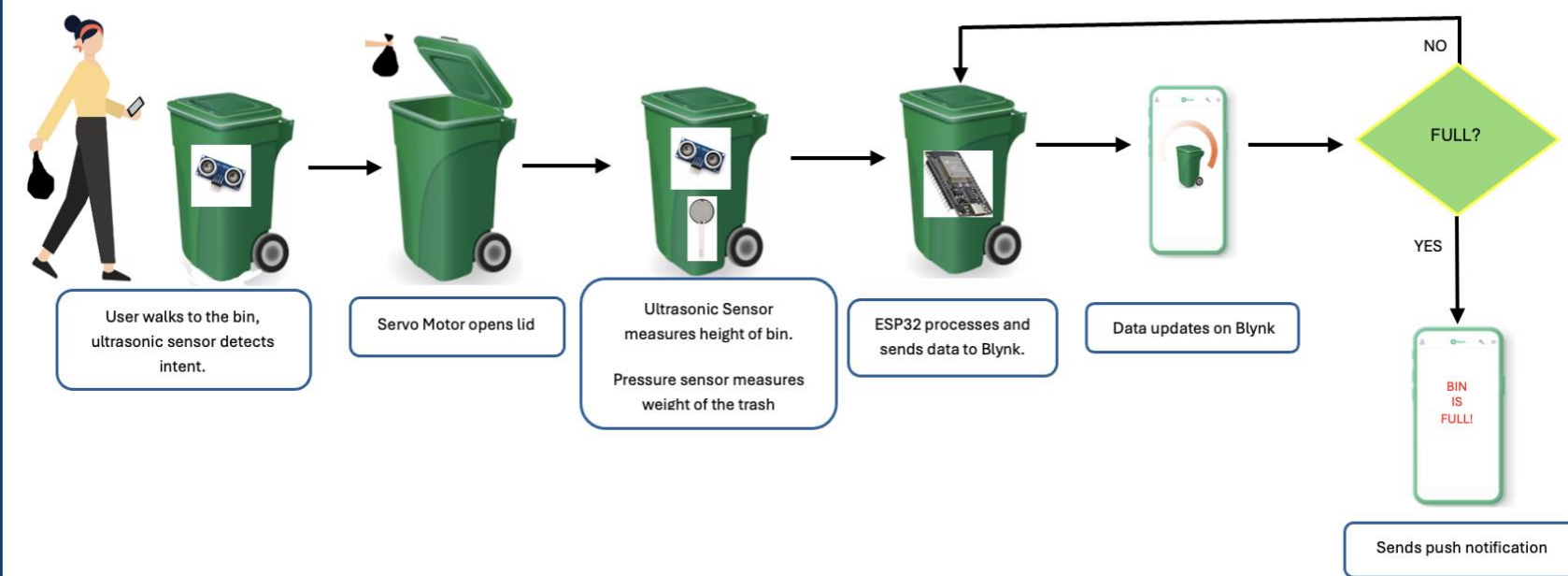


Component & How It Works

1. **ESP32 Microcontroller:** controls the bin and connects to Wi-Fi to send alerts when the bin is full.
2. **Ultrasonic Sensor (HC-SR04):** measures the bin's fill level, and ensures the lid opens only when someone intends to throw waste.
3. **Servo Motor:** opens and closes the lid.
4. **Pressure Sensor:** to measure the weight of the waste.
5. **Battery:** the power source for the smart bin.



Key Concept



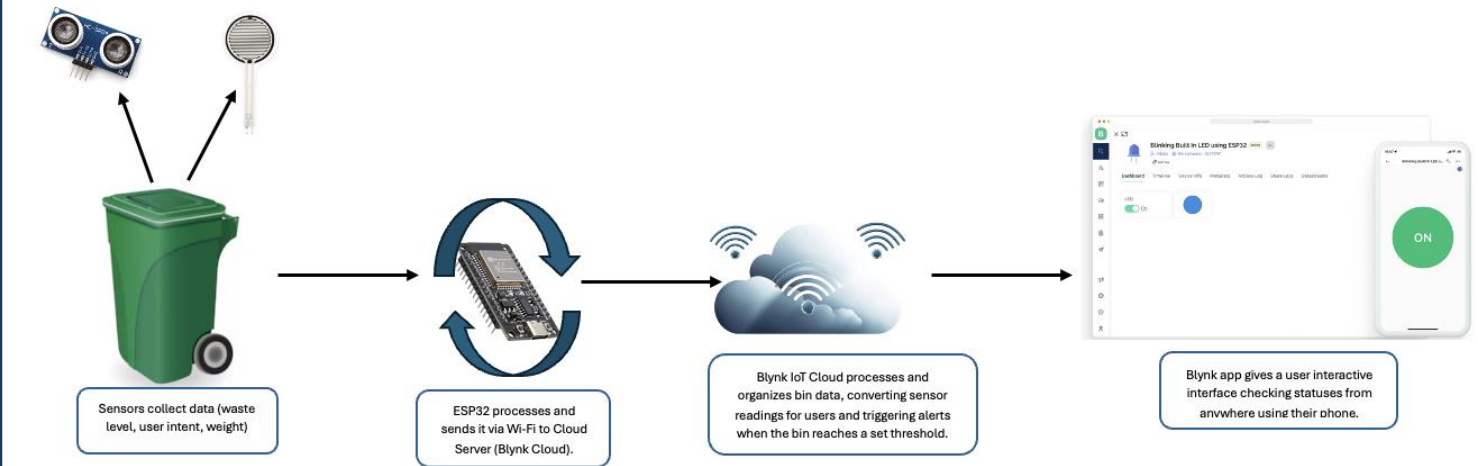
Methodology

This project follows a step-by-step process that includes planning, hardware selection, circuit development, programming, and testing.

1. **System Design & Planning:** define how the bin functions, including waste level measurement, alert notifications, and automated lid control ensuring energy efficiency.
2. **Selection of Component:** choose the required hardware, including bin with lid, ESP32, ultrasonic sensors, pressure sensor, servo motor, and battery.
3. **Circuit Development & Prototyping:** At this stage, a breadboard and jumper wires should be used to set up the circuit for initial testing before implementation. Connections include sensors and ESP32 to the microcontroller for data collection, servo motor to the microcontroller for lid movement.
4. **Programming & Implementation:** Develop and refine the ESP32 code to measure fill levels, trigger alerts, open/close the lid, and optimize system responsiveness. Close the lid after a short delay. The code is tested and improved to maintain accuracy and responsiveness.
5. **Testing & Optimization:** Conduct tests to verify sensor accuracy, lid mechanism, power efficiency, and ensure Blynk alerts work correctly when the bin reaches capacity. At this stage, continuous testing is done to detect errors. Based on results, settings are adjusted constantly to improve accuracy, and responsiveness.
6. **Final Build & Deployment:** After successful testing and all requirements are met, all components are securely placed inside the bin. Temporary connections are replaced with durable wiring or soldering, and the final product is designed for practical use in homes, events, and public spaces.

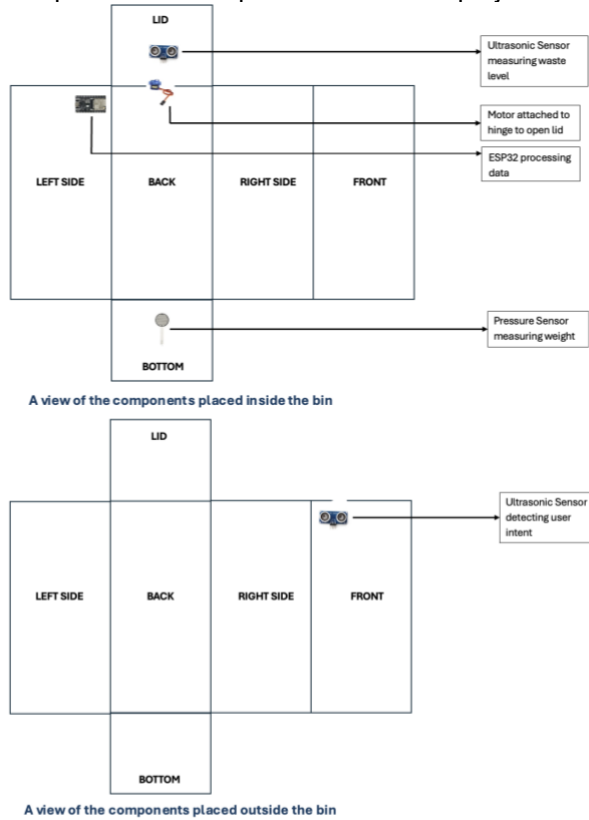
Network Architecture

The bin uses Wi-Fi to send real-time waste level updates, making it easier to monitor and manage waste collection efficiently. The system connects to Blynk, allowing users to track bin status remotely through a mobile app.



Results

The Smart Waste Bin is a practical solution tailored for modern public spaces, shopping malls, shared accommodations, commercial buildings, and indoor social events, especially for garbage management agencies. By monitoring trash levels, preventing unnecessary lid openings, and sending alerts when full, this project supports cleaner venues, improved hygiene, and more efficient waste disposal practices. With further enhancements, this technology can revolutionize waste management in large-scale gatherings and entertainment industries. Below are views of where the components will be placed to fulfil this project.



Recommendations

1. Implement AI-powered image recognition to differentiate between waste types, and so it can perform waste sorting.
2. Integrate a solar panel to support sustainable operation.
3. Equip the bin with a GPS module to help waste collection agencies track bin locations.

References

1. Orimogunje, A. M., Fred-Ahmadu, O. V., Adewale, A. A., & Olaitan, A. (2022). *Design and development of smart waste bin for effective waste collection and management*.
2. Shen, N. J., et al. (2020). *Design and development of smart waste management system. International Journal of Advanced Trends in Computer Science and Engineering*, 9(5), 6495-6502